Candidate Seat no……………….

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Third Semester of B. Tech (EC) Theory Examination**

**March-Apr 2018**

**EC202 Network Theory**

**Date: 30/03/2018, Friday** **Time: 01:30 pm To 04:30 pm** **Maximum Marks: 70**

***Instructions:***

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION – I

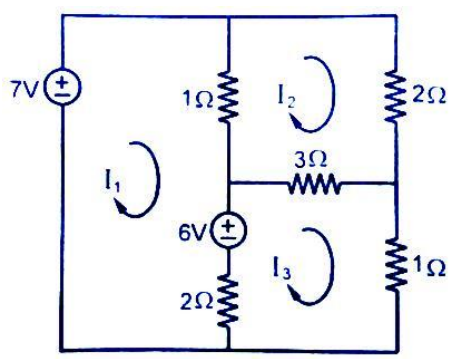
**Q:1 Answer the following** **[07]**

**a.** State the difference between circuit and network. [02]

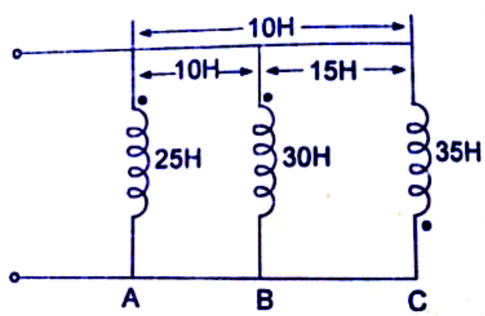
**b.** Explain different types of network. [05]

**Q:2 Attempt any three** **[18]**

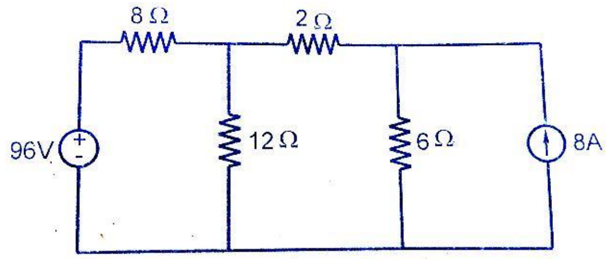
**a.** Determine the currents $I_1$, $I_2$ and $I_3$ in the given network using mesh analysis.



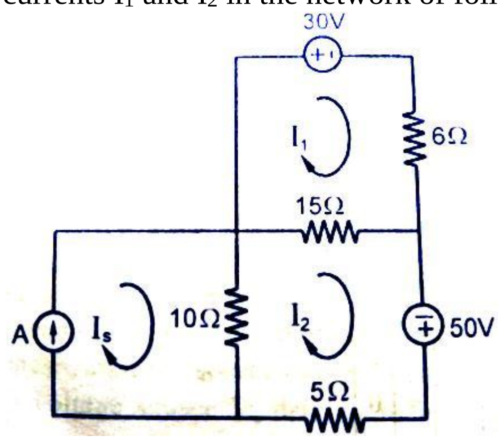

**b.** Determine the inductance between the terminals for a three coil shown in following figure.



**c.** Find the voltage across 6 Ω reststor using node analysis in the following network.

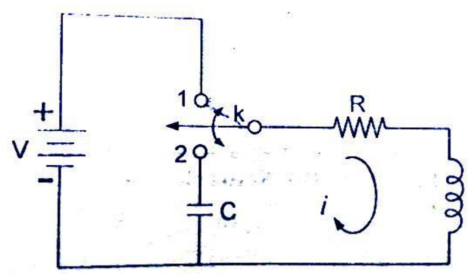



**d.** Determine the mesh currents $I_1$ and $I_2$ in the network of following figure.



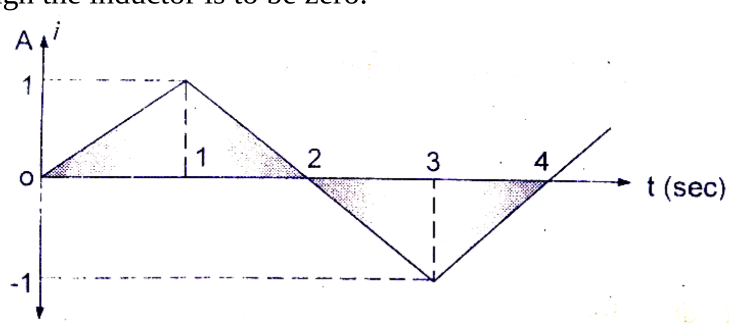

**Q:3 Attempt any two** **[10]**

**a.** In the netwrok of following figure, switch K is changed from 1 to 2 at t=0.Find the values of i, di/dt and $d^2i/dt^2$ at t=0+.Here, R=1000Ω,L=1H,C=0.1µF and V=100V.

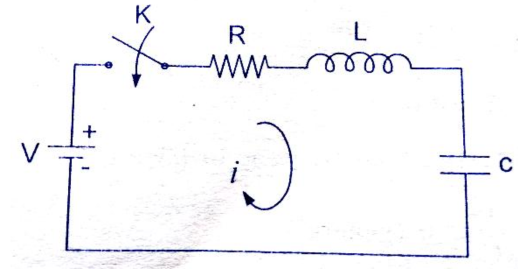



**b.** A current source having wave form shown in following figure is connected across the terminals of an inductor of 10 mH. Show

1) The voltage waveform,
2) The charge wave form through the device. Assume the initial condition through the inductor is to be zero.



**c.** In the network of following figure,the switch K is closed at t=0 with capacitor uncharged and with zero current in the inductor. Find the values if i , di/dt and $d^2 i/dt^2$ at t=0+. Here, V=100V, R=10Ω, L=1H and C=10μF.



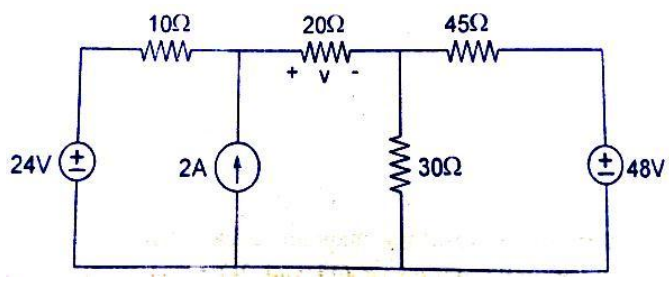

## SECTION – II

**Q:4** Obtain the particular solution for R-C transient circuit. **[07]**

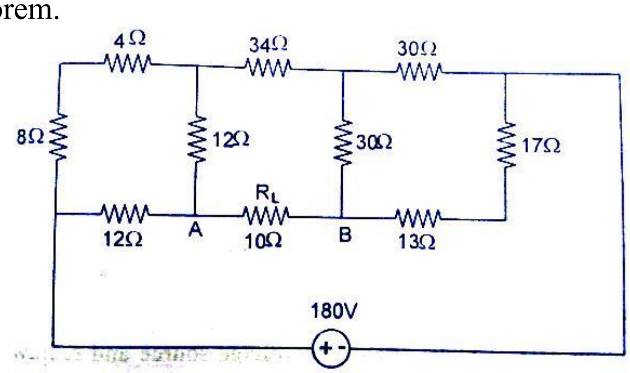
**Q:5** **Attempt any three.** **[18]**

**a.** Explain and analyze the response of series R-L-C circuit as related to the S plane location of roots.

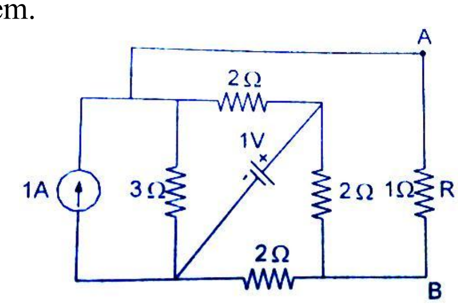
**b.** Use superposition theorem to find value of 'V' across 20Ω resistor in the following network.




**c.** Find the current in 10Ω resistor in the network of following figure using thevenin's theorem.

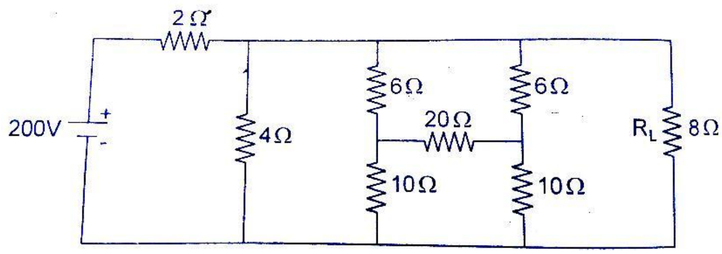



**d.** Determine the current in R=1 Ω resistor of network shown in following figure using Norton's theorem.

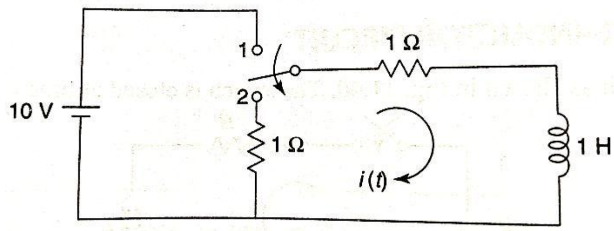


**Q:6 Attempt any two. [10]**

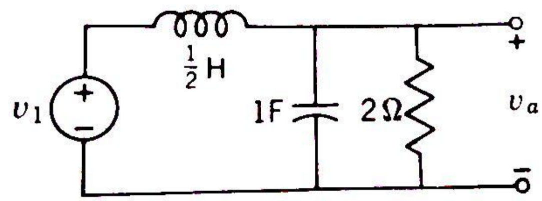
**a.** Find the current thorugh 8Ω resistor using Norton's theorem.



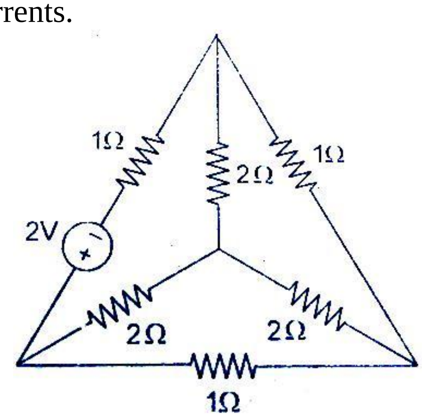

**b.** In the network of following figure, the switch K is moved from position 1 to position 2 at t=0(a steady state existing in position 1). Solve for the current i(t) for t>0 using laplace transformation method




**c.** For the following given network, find Va(t) in the steady state if $V_1$=2 sin 2t.




**d.** For the network shown in following figure,write down the tieset matrix and obtain the equilibrium equation in matrix form using KVL.Calculate the loop currents and branch currents.

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